

(DRAFT) Republican Media Diets and Intraparty Realignment Post-January 6

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BACKGROUND

Two months after his defeat in the 2020 presidential election, on January 6, 2021, thousands of then-president Donald Trump's supporters flocked to the U.S. Capitol building. They did so with the goal of prohibiting a peaceful transition of power to Joe Biden, the president-elect. In the days following the November presidential election, Mr. Trump had propagated and fueled claims to his base that the election had been rigged in favor of Mr. Biden. Emboldened by Mr. Trump's repeated claims of election tampering, the rioters sought to prevent a joint session of Congress in which electoral votes from the 2020 presidential election were being certified. Beforehand, Mr. Trump had held a "Save America" rally in proximity to the White House, where he continued to repeat unevidenced claims that the election had been stolen by the Democrats, giving an impetus to the protesters. Soon, the riot turned violent and destructive through security breaches, conflicts with law enforcement and vandalism of the Capitol itself, a historic monument and symbol of democracy that had not been attacked in such a nature since the War of 1812 (Krakowski & Morales, 2025). Mr. Trump's role in stoking the flames of discontent post-election, including his behavior at the "Save America" rally as well as his hesitancy to denounce the rioters' destructive actions, made him culpable or at least partially responsible for the attack in the eyes of many.

As a unique form of electoral political violence, the insurrection was initially decried by Democrats and Republicans alike; however, in the weeks that followed, certain Republicans resisted Congressional processes to hold Mr. Trump accountable for the January 6 events (Fandos, 2021; Krakowski & Morales, 2025). For some Republicans, the events of January 6 represented a shift in the GOP toward a party that condoned political violence and disrespected democratic processes, and therefore, was no longer representative of their beliefs. For others, despite Mr. Trump's cemented position as the leader of this populist and extreme shift, maintaining their party affiliation reigned supreme. Therefore, the aftermath of the January 6 events was characterized by "subsequent struggles within the Republican Party over how to respond to the insurrection", with such tension indicating the potential beginnings of intraparty realignment (Eady et al., 2023, p. 1152). In the tumultuous days that followed, news coverage across media outlets greatly varied, with certain organizations emphasizing the outrageousness of the crisis while others downplayed and justified the events. It follows that media outlets with an ideological slant toward the right might have reported more sympathetically toward Mr. Trump and his protesters' actions. While many factors indeed contributed to Republican decisions to align or de-align themselves with Trump as an extreme figure of the GOP, with this context in mind, analyzing media diets provides insight into the mechanisms related to party identification in the aftermath of a violent political event. In particular, reliance on mainstream versus fringe news outlets may prove interesting in understanding how media systems are associated with support for populist leaders of the far right, such as Mr. Trump.

Therefore, this study seeks to explore how media diets among Republicans are associated with support for Trump as a continuing national political figure for years to come. Republicans that choose to support Trump as a political figure in the future are understood to continue aligning themselves with the Republican party, shifting toward its new extreme.

Republicans who choose not to support Mr. Trump as a continuing political figure are understood to be shifting away from the GOP, indicating a potential realignment within the party itself.

Using data from the Pew Research Center's American Trends Panel, this study uses logistic regression models to determine the odds of future Trump support among Republicans, with media diets (between mainstream and fringe outlets) as a key independent variable. Preliminary results indicate that Republicans who rely on mainstream and fringe conservative media are the most likely within their party to support Mr. Trump as a future political figure. Republicans who consume only fringe content or only mainstream content are not statistically different from one another. However, Republicans who do consume neither fringe nor mainstream conservative content are significantly less likely to support Trump in the future, indicating that media intensity, rather than type, may be a guiding driver in the relationship between media diet and intraparty GOP dynamics.

This draft is structured as follows. First, a review of relevant literature is presented, followed by the hypotheses explored in the study. Next, the methodology and datasets utilized are discussed. Preliminary results are shown, followed by an acknowledgement of this study's current limitations.

LITERATURE

According to the literature, Americans are more polarized than ever (Iyengar et al., 2019; Groenendyk, 2018). Studies within the past decade have established time and time again that "ordinary Americans", in addition to political elites, increasingly dislike and distrust those from the other party", claiming that now, more than ever, partisanship is a uniquely salient feature of Americans' social identities (Iyengar et al., 2019, p. 129; Iyengar & Krupenkin, 2018; Abramowitz & Webster, 2016). Ideological divides between Democrats and Republicans have only grown since the beginning of the 21st century, with a rise of "negative partisanship" being observed in the electorate (Abramowitz & Webster, 2016, p. 12). Existing studies, using American National Election Studies (ANES) data, reinforce this phenomenon, with recent elections being characterized by higher-than-ever levels of party loyalty and straight-ticket voting (Abramowitz & Webster, 2016, p. 12-13, Groenendyk, 2018).

While it is well-documented that partisan behavior is on the rise, interestingly, since 2016, an increasing share of U.S. voters have claimed to identify as Independents instead, with party loyalty being informed more by negativity toward the opposing party rather than positive feelings about one's own party (Abramowitz & Webster, 2016, p. 12-16). Subsequent studies have reaffirmed this notion, suggesting that partisans operate through a "lesser of two evils" approach to rationalize continuing their current party affiliation (Groenendyk, 2018, p. 159). Therefore, partisans in both the Republican and Democratic parties are "reporting less positive evaluations of, and greater ambivalence toward, their own party" (Groenendyk, 2018, p. 159-163). This counters previous implications of mass parties being cohesive and homogeneous entities. Given this trend of rising ambivalence towards one's own in-group, could it be possible that party identity is becoming more malleable? If it is well established

that polarization and partisanship have been on the rise between the Democratic and Republican parties, then how has intraparty affect continued to evolve since the rise of the populist right in the United States in 2016?

The current literature has focused on reinforcing the fact that inter-party polarization is currently extremely strong in the United States, leading to a rise in partisanship. However, intraparty dynamics are comparatively less explored, particularly in the context of political violence events. When an act of political violence is perpetrated by a member of an opposing ideological group, it follows that the tendency is for one to feel less warmly toward the opposing party, as they have engaged in non-democratic behavior (Zyla, 2024). Similarly, when the target of an act of political violence is a member of the ideological in-group, individuals tend to prefer harsher penalties for the perpetrator, as “violence against a copartisan...might cue psychological systems that operate as though the violence exists in one’s immediate environment” (Zyla, 2024, p. 459). However, when the perpetrator is of the ideological in-group, the dynamics of party identity start to become hazy. For example, using X (formerly Twitter) data, one relevant study regarding the events of January 6, 2021, found that escalations of political violence did indeed reduce support for leaders associated with the perpetrators (Krakowski & Morales, 2025). In this case, Republican politicians who were strongly associated with Trump, who is “commonly credited with the incitement of the Capitol riot”, experienced a significant drop in online support in the short-term aftermath of the insurrection (Krakowski & Morales, 2025, p. 1488). Emerging within the literature on political violence is its effects on political behavior and party identity, as exemplified by a 2023 study that found that, following the January 6 events, a large-scale de-identification of Republican party affiliation occurred among social media users - an effect that persisted in the long term (Eady et al., 2023). However, as public de-identification does not necessarily imply a true change in political behavior or beliefs, more work is necessary to determine whether partisan identities within the Republican party did indeed shift during this time. The present study, therefore, seeks to explore this phenomenon further to investigate whether the seeds of intra-party realignment occurred in the aftermath of a violent political act widely accepted, at varying degrees, to be due to the actions of an ideological in-group member.

Citizens and elites alike typically denounce political violence as an abstract concept, arguing that it erodes democratic stability by paving the way for violence to be an acceptable political tool (Zyla, 2024, p. 1481). The insurrection at the Capitol on January 6, 2021 has been described as possibly “the most dramatic contemporary manifestation of deep political polarization in the United States” in its history, representing a “significant case of political violence with electoral motivations” (Eady et al., 2023, p. 1151; Krakowski & Morales, 2025, p. 1482).

Political violence as a tool of suppression, control, and manifestation of political dissent is understudied in the context of a “mature democracy, where established electoral processes [and] effective accountability mechanisms”, in theory, are meant to quell episodes of political violence and its appeal (Krakowski & Morales, 2025). The peculiarity of the Capitol insurrection in the United States, therefore, brings up questions of polarization and how it manifested to such a degree that such a violent event could occur. Analysis of news media consumption habits is one avenue within many potential ones into exploring how

support for political leaders implicated in events of political violence evolves in the aftermath of such broadly visible and shocking acts.

Media and news outlets in the United States have experienced two major shifts regarding the priority of neutral reporting (Levendusky, 2013). Around the beginning of the 20th century, news outlets began to emphasize neutrality rather than taking an explicit ideological stance in their coverage (Levendusky, 2013). This trend continued into the 21st century, but of recent, with the rise of new forms of media and the rise of cable news as competitors to network news, “partisan media outlets have reemerged in the U.S.” (Levendusky, 2013, p. 565; Prior, 2007; Peterson et al., 2021). This has led to an increasingly polarized media landscape, in which outlets may selectively determine which topics are newsworthy, which issues should be emphasized, and the ideological slant (if any) that should be used while discussing and analyzing said issues. Through these choices, media outlets privilege certain frames over others, taking advantage of viewers’ predisposed (political) attitudes, helping them “make sense of the world” while moving past the simple task of purely conveying information (Rosensteil, 2006; Levendusky, 2013, p. 566). Different framing routes allow news outlets the possibility to “[obscure] the connections between political problems and the actions or inactions of political leaders...” therefore “[weakening] the accountability of elected officials” (Iyengar, 1996, p. 59).

Manipulation of media framings also allows consumers to select information exposure that aligns more closely with their preexisting beliefs, thus creating a reinforcement loop (Peterson et al., 2021; Levendusky, 2013). This partisan divide is particularly stark regarding election-related news compared to nonpolitical news (Peterson et al., 2021). It is important to note that this relationship is bidirectional; media framings may both reflect increased partisanship and polarization while also influencing their rise in the United States (Levendusky, 2013). While it holds that few news media organizations in the United States are truly neutral in their reporting, mainstream and fringe outlets make use of frames differently. Mainstream media do make attempts to favor objectivity, while fringe outlets do no such thing (Jamieson et al., 2007; Jamieson and Cappella, 2008; Levendusky, 2013). In these outlets’ news coverage, curated facts that demonstrate issues as a mere matter of inter-party tensions “[cue] voters that they should understand these issues through the lens of partisanship” rather than neutrality, thus reinforcing in- and out-group biases (Price, 1989; Yzerbyt et al., 2006; Levendusky, 2013, p. 567).

In the context of the populist right reaching new heights of electoral power, as exemplified by the election of Donald Trump in 2016 and his continued presence as an extreme political figure within the GOP, the increasingly polarized media ecosystem is particularly relevant. Populists, from both the supply and demand sides, have unique relationships to media systems. Populist leaders often denounce traditional media sources, proclaiming them to be instruments of the ‘establishment’ (Mudde, 2007, p. 67). In particular, populist politicians on the right frequently attack mainstream journalists as “leftists, liars, and traitors” (Mudde, 2007, p. 67). Fringe outlets that are unrestricted by traditional journalistic reporting standards of neutrality are, therefore, particularly useful for spreading populist ideas, making them more appealing to those who already support populist candidates. Regarding the current study, partisan news reporting is particularly relevant in the context of the 2020 presidential election, as “partisanship is more pronounced when considering

campaign coverage”— consumers are expected to be even more selective with their news consumption during times of high political stress, such as elections (Peterson et al., 2021, p. 2). With the development of such a polarized media ecosystem, studies in the literature have aimed to test whether watching and engaging with partisan media sources does indeed increase negative affect toward opposing parties (Levendusky, 2013; Peterson et al., 2021). For example, a series of experiments by Levendusky (2013) during the 2016 U.S. presidential campaign period, a time of heightening polarization in the United States, demonstrated that viewers of partisan media outlets do indeed “perceive the other party more negatively...trust them less, [and are] less supportive of bipartisanship”, thus reinforcing blame patterns that scapegoat the opposing party rather than one’s own (Levendusky, 2013, p. 565).

The literature has, therefore, extensively examined the effects of ideologically-like media consumption on citizens’ political beliefs and attitudes. It has also begun to observe what happens when subjects consume what is called cross-cutting partisan media — in other words, media that resists or fully goes against their current ideological beliefs (Levendusky, 2013). Current studies show that, logically, while ideologically like content causes less warm feelings toward the opposing party, cross-cutting content seems to “have little effect on either same-party or opposite-party feeling thermometers” (Levendusky, 2013, p. 571). The literature has not yet examined if these associations remain similar when in the context of a major act of political violence facilitated by, or associated with, an ideological in-group member. Therefore, this study aims to take this exploration further by examining intraparty dynamics within the Republican Party, particularly in the aftermath of a major act of political violence that generated significant controversy and was potentially disruptive to the party’s dynamics.

With this context in mind, this article addresses the following guiding research questions:

RQ: Among Republicans, how did differences in media consumption habits relate to intentions to support Trump as a political leader in the future in the aftermath of the January 6 Capitol attack? Is there a marked difference in future support for Trump between those who consume mainstream conservative media and those who consume fringe conservative media?

Testing these hypotheses will provide insights into how distinct informational environments provide a perspective into what may be the beginning of separate constituencies forming within the Republican Party — the seeds of intraparty realignment.

H1: Republicans who rely on ideologically-like (conservative) news content are more likely to continue supporting Trump as a future political figure than those who do not consume conservative media content.

H1b: Republicans who consume fringe conservative news content are the most likely to continue supporting Trump as a future political figure, compared to those who only consume mainstream conservative content.

H1, this study's broader hypothesis, reflects the nature of seeking and consuming ideologically consistent news content, therefore assuming that conservative media outlets, regardless of whether they are mainstream or fringe organizations, have consistently portrayed Trump positively and framed criticisms of his actions as partisan attacks.

H1b adds another layer to the previous hypothesis, expecting that fringe outlets such as talk radio often adopt more populist and anti-institutionalist rhetoric than mainstream sources.

METHODOLOGY¹

In order to test the previously posited hypotheses, this paper makes use of public opinion survey data from the American Trends Panel (ATP) run by the Pew Research Center, a nonpartisan American think tank headquartered in Washington, D.C. The ATP is based on a random sample of individuals surveyed across the United States across multiple waves, allowing long-term trends such as media diet to be captured. Most notably, the ATP was relevant for this study because of its survey timing, which captured respondent attitudes nearly immediately after the events of January 6, 2021. The waves being used for this study were Waves 79 and 80.

Wave 79 captured respondents' media diets prior to the election. As media diets are likely formed over time, using data for this variable that was captured prior to the events of January 6 allowed for a better picture of each respondent's general conservative media diet. Questions about media consumption in this wave focused on asking respondents whether certain mainstream sources, such as Fox News, CNN, MSNBC, and other outlets, including talk radio and fringe media sites, had provided the respondent with news about the presidential election since polls closed on Election Day. The questionnaire for Wave 79 was conducted from November 18 to November 29, 2020, thus capturing media consumption habits nearly right after the 2020 polls closed. Classification of outlets as 'conservative' is based on the Pew Research Center's 2020 report on U.S. Media Polarization (Pew Research Center, 2020).

From Wave 80, demographic variables including age, gender, income, and education categories were pulled. Additionally, information on respondent ideological stance (from very liberal to very conservative) was taken from this wave. This wave also provided information on party identification (Republican, Democrat, or party lean). Exposure to news about the events of January 6 was controlled for, as the variable for news consumption alone does not explain what kinds of news the respondents consumed. A variable measuring whether respondents believe Trump bears responsibility for the events of January 6 was also included as a control; a model that did not control for this would risk confoundment.

Lastly, the dependent variable - support for Trump's future as a political figure in the United States - was included. As Trump's association with the incitement of the January 6 events marked him as a more extreme figure within the Republican party, to analyze whether the beginnings of intraparty realignment are occurring, this is a key variable. Republican

¹ More information on variable choice, model iterations, as well as diagnostics will be provided in the final draft.

respondents who continued to support Trump as a political figure in the future were more likely to resist a split in the GOP, while those who de-identified with Trump are assumed to indicate a stronger tendency towards realignment.

To test the previously posited hypotheses, logistic regression models were iterated, with the results of the final model being reported below. Unlike Ordinary Least Squares (OLS) models, Logistic regression models allow for an analysis of binary outcome variables, making this a suitable method for testing the binary dependent variable (support for Trump as a political figure versus non-support). As logistic regression models do not, however, allow for causal claims, it is imperative that estimates must only be interpreted as associations. The final model was found to correctly predict 74.9% of the responses.

RESULTS²

Table 1 reports the log-odds coefficients for the final model.

Logistic Regression Results	
	<i>Dependent variable:</i>
	Support Trump in Future
Fringe Type Only	-0.023 (0.269)
Fringe and Mainstream	0.587*** (0.139)
Neither Media Type	-0.432** (0.174)
Ideology (Higher = More Conservative)	0.403*** (0.081)
Exposure to Jan 6 Information	0.014 (0.113)
Assigns Trump Responsibility	-1.881*** (0.096)
Gender (1 = Male)	0.449*** (0.123)
Age	0.004 (0.061)
Education	-0.113*** (0.043)
Income	-0.102*** (0.022)
Constant	2.649*** (0.467)
Observations	1,883
Log Likelihood	-870.959
Akaike Inf. Crit.	1,763.917
<i>Note:</i>	* p<0.1; ** p<0.05; *** p<0.01

Table 1: Log-Odds Coefficient Estimates of Final Model

² Further analysis of the results will be provided in the final article.

For ease of interpretation, **Table 2** reports the odds-ratios of the final model.

	OR	2.5 %	97.5 %
(Intercept)	14.138	5.719	35.646
Fringe Type Only	0.977	0.581	1.671
Fringe and Mainstream	1.799	1.370	2.364
Neither Media Type	0.649	0.461	0.912
Ideology (Higher = More Conservative)	1.496	1.278	1.755
Exposure to Jan 6 Information	1.014	0.811	1.266
Assigns Trump Responsibility	0.153	0.126	0.183
Gender (1 = Male)	1.567	1.233	1.996
Age	1.004	0.891	1.130
Education	0.893	0.821	0.971
Income	0.903	0.865	0.943

Table 2: Odds-Ratios and 95% Confidence Intervals of Final Model

The media consumption variable treats mainstream conservative media-only as the baseline. Fringe conservative media consumption was not found to have a significant difference in effect on future support for Trump than conservatives who only consume mainstream media. However, the estimate of consuming both mainstream as well as fringe media was highly significant ($p < 0.001$), positive, and strong, indicating that conservatives who consumed both types of media were approximately 79% more likely to support Trump's future as a political figure than conservatives who only consume mainstream media.

The estimated coefficient for a media environment that included neither mainstream conservative media nor fringe conservative media was negative, strong, and significant ($p < 0.01$), indicating that those who fall into this group were roughly 35.1% less likely to support Trump in the future than solely mainstream consumers. These results indicate that respondents who were embedded in both mainstream and fringe conservative media are the strongest Trump loyalists, and therefore more resistant to tendencies toward Republican intraparty realignment. As Republicans who are disengaged from partisan conservative media

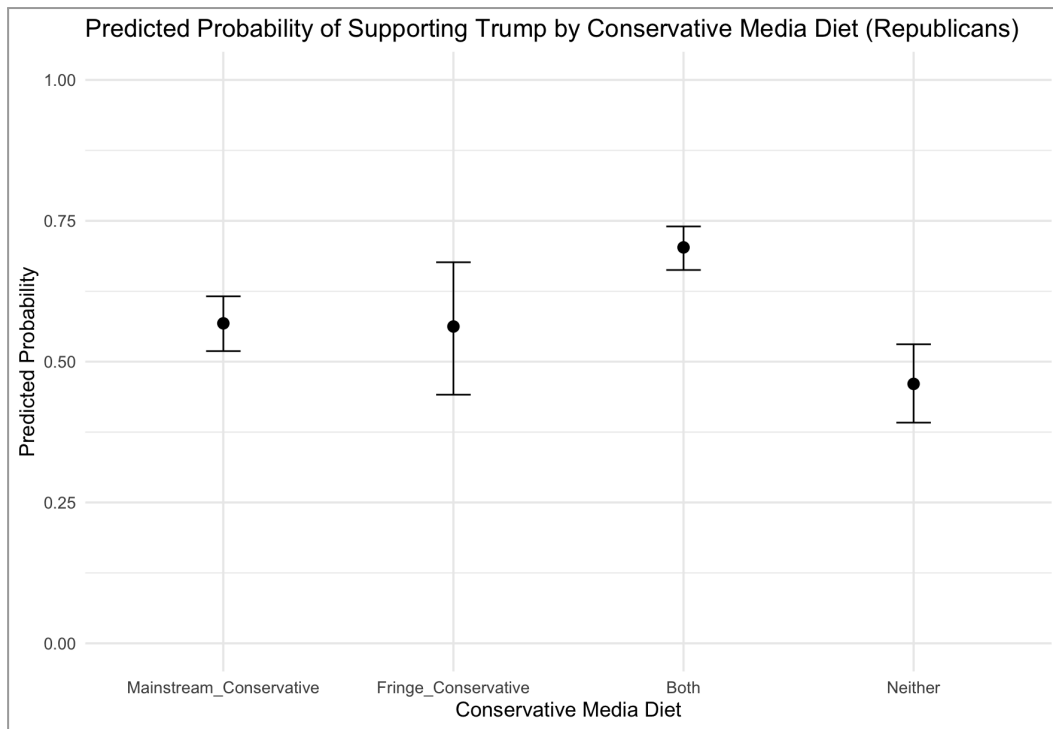


Figure 1: Predicted Probability of Supporting Trump's Political Future

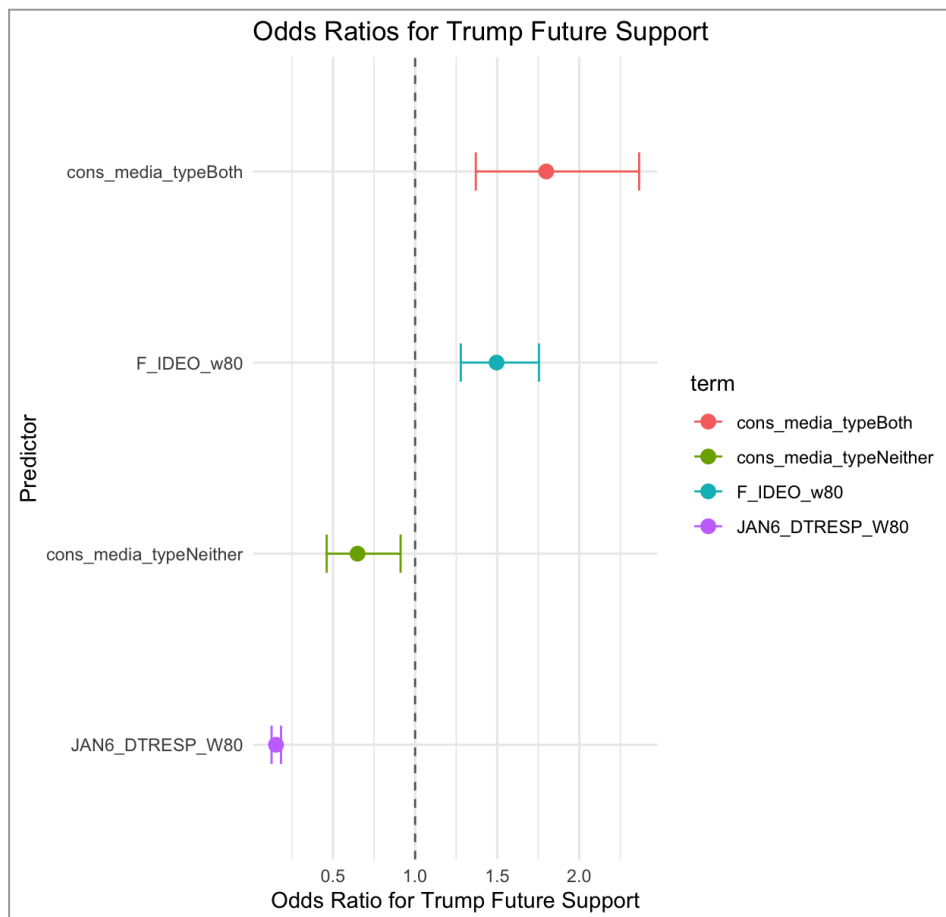


Figure 2: Odds-Ratios of Supporting Trump's Political Future

are significantly less loyal to Trump, and fringe-only users are not statistically distinct from mainstream-only users, these results further suggest that media intensity, rather than fringe-ness alone, matters. Therefore, **H1** is supported by these results, while **H1b** is rejected.

Certain other attitudinal predictors were also statistically significant. Unsurprisingly, ideology was strongly and positively significant ($p < 0.001$), indicating for every one unit increase in conservatism, Republicans are about 49.6% more supportive of Trump's future as a political figure. This further points toward Trump moving into place as a more far-right, extreme figurehead of the Republican party. The variable measuring perceptions of Trump's responsibility for the events of January 6 was significant ($p < 0.001$), strong, and negative, indicating that attributing blame to Trump is associated with approximately an 84.7% decrease in odds of supporting his political future, even among members of his party. Hearing about January 6 was not a statistically significant predictor in explaining support for Trump's political future.

These results show that partisan media intensity may indeed inform intraparty dynamics of split support for a political elite; accountability perceptions also form a critical divide within the GOP. Thus, media consumption and informational divides do play a role in informing future support for Trump; however, acceptance of Trump's responsibility for the January 6 insurrection remains the primary point of fracture within the party. The present results do indicate the beginnings of intraparty realignment.

LIMITATIONS

In conclusion, it is necessary to acknowledge the limitations of this study. Firstly, given that the data is derived from survey responses, social desirability bias may play a role. Particularly in the recent aftermath of January 6, of which the events were decried by numerous groups and entities, respondents may be reluctant to divulge their true political beliefs and sources of news out of shame.

Additionally, when listing different types of news content, the survey question measuring media outlet reliance only provided limited examples of each type of source. Therefore, respondents may have believed they could only indicate that they used that type of media if they used the exact outlets listed as examples in the question, leading to inaccurate media use reporting. The subsequent classification of these media types into "fringe" and "mainstream" may also lose nuance; not all mainstream media organizations nor fringe outlets report the same way.

Thirdly, an important methodological limitation is that the estimates generated by the logistic regression model only represent associations, and not causation. It is indeed possible that support for Trump's political future also informs the media that respondents choose to consume, and therefore, these results should be interpreted as relationships rather than determined causal directions.

Lastly, it is essential to note that the results of this study only reflect short-term, immediate attitudes, rather than long-term effects. It is possible that Republicans who, in the initial weeks after January 6, 2021, felt a disconnect from the Republican party, later

re-identified or re-solidified their alignment with the in-group so as to not lose benefits from membership (Krakowski & Morales, 2025).

REFERENCES

- Abramowitz, A. I., & Webster, S. (2016). The rise of negative partisanship and the nationalization of US elections in the 21st century. *Electoral Studies*, 41, 12-22.
- Eady, G. et al. (2023). Do violent protests affect expressions of party identity? Evidence from the capitol insurrection. *American political science review*, 117(3), 1151-1157.
- Fandos, N. (2021). House backs Jan. 6 commission, but Senate path dims. The New York Times, 19 May. Available at:
<https://www.nytimes.com/2021/05/19/us/politics/housejan-6-commission.html>
- Groenendyk, E. (2018). Competing motives in a polarized electorate: Political responsiveness, identity defensiveness, and the rise of partisan antipathy. *Political Psychology*, 39, 159-171.
- Groenendyk, E. et al. (2020). Intra party polarization in American politics. *The Journal of Politics*, 82(4), 1616-1620.
- Iyengar, S. (1996). Framing responsibility for political issues. *The annals of the American academy of political and social science*, 546(1), 59-70.
- Iyengar, S., & Krupenkin, M. (2018). "Partisanship as Social Identity: Implications for the Study of Party Polarization." *Forum* 16 (1): 23–46.
- Iyengar, S. et al. (2019). The origins and consequences of affective polarization in the United States. *Annual review of political science*, 22(1), 129-146.
- Jamieson, K. H. et al. (2007). The effectiveness of the press in serving the needs of American democracy. *Institutions of American democracy: A republic divided*, (717), 21-51.
- Jamieson, K. H., & Cappella, J. N. (2008). *Echo chamber: Rush Limbaugh and the conservative media establishment*. Oxford University Press.
- Krakowski, K., & Morales, J. S. (2025). Does political violence backfire in mature democracies? Evidence from the Capitol insurrection in the USA. *Journal of Peace Research*, 62(5), 1479-1497.
- Levendusky, M. (2013). Partisan media exposure and attitudes toward the opposition. *Political communication*, 30(4), 565-581.
- Mudde, C. (2007). *Populist radical right parties in Europe*. 1st edition. [Online]. Cambridge: Cambridge University Press.

- Peterson, E., Goel, S., & Iyengar, S. (2021). Partisan selective exposure in online news consumption: Evidence from the 2016 presidential campaign. *Political science research and methods*, 9(2), 242-258.
- Pew Research Center. (2020). "U.S. Media Polarization and the 2020 Election: A Nation Divided"
- Price, V. (1989). Social identification and public opinion: Effects of communicating group conflict. *Public Opinion Quarterly*, 53(2), 197-224.
- Prior, M. (2007). *Post-broadcast democracy: How media choice increases inequality in political involvement and polarizes elections*. Cambridge University Press.
- Rosensteil, T. (2006). How the mass media divides us—Two alternative perspectives. *Red and blue nation*, 1, 249-54.
- Yzerbyt, V. Y. et al. (2006). *Social comparison and group-based emotions* (pp. 174-205). Cambridge, UK: Cambridge University Press.
- Zyla, J. M. (2024). Partisan Bias in Episodes of Political Violence. *American Politics Research*, 52(4), 451-462.